Experiment 8.1

Simulation of EMI Sources and Interference Analysis using MATLAB

Name: Rohith Krishna R	**Reg. No:** 192312013

Aim

To simulate different Electromagnetic Interference (EMI) sources (sinusoidal, broadband and impulse), analyse their effect on a desired signal, and evaluate the improvement in SNR after filtering.

Software Used

MATLAB

Theory / Formulas

Received signal = Desired signal + EMI signal.
SNR (dB) = $10 \log_{10}(P_s / P_n)$, where P_s = mean(signal^2) and P_n = mean(noise^2).
A low-pass filter is applied to the received signal to suppress the EMI, and the SNR is recomputed using the residual error (filtered - clean signal).

Procedure

1. Generate the desired signal and the EMI sources (sinusoidal, broadband noise, impulse).
2. Add the EMI to the desired signal to form the received signal.
3. Compute the frequency spectrum using FFT.
4. Calculate SNR before filtering.
5. Apply a low-pass filter and calculate SNR after filtering.
6. Plot the time-domain signals, spectrum and SNR comparison.

Output

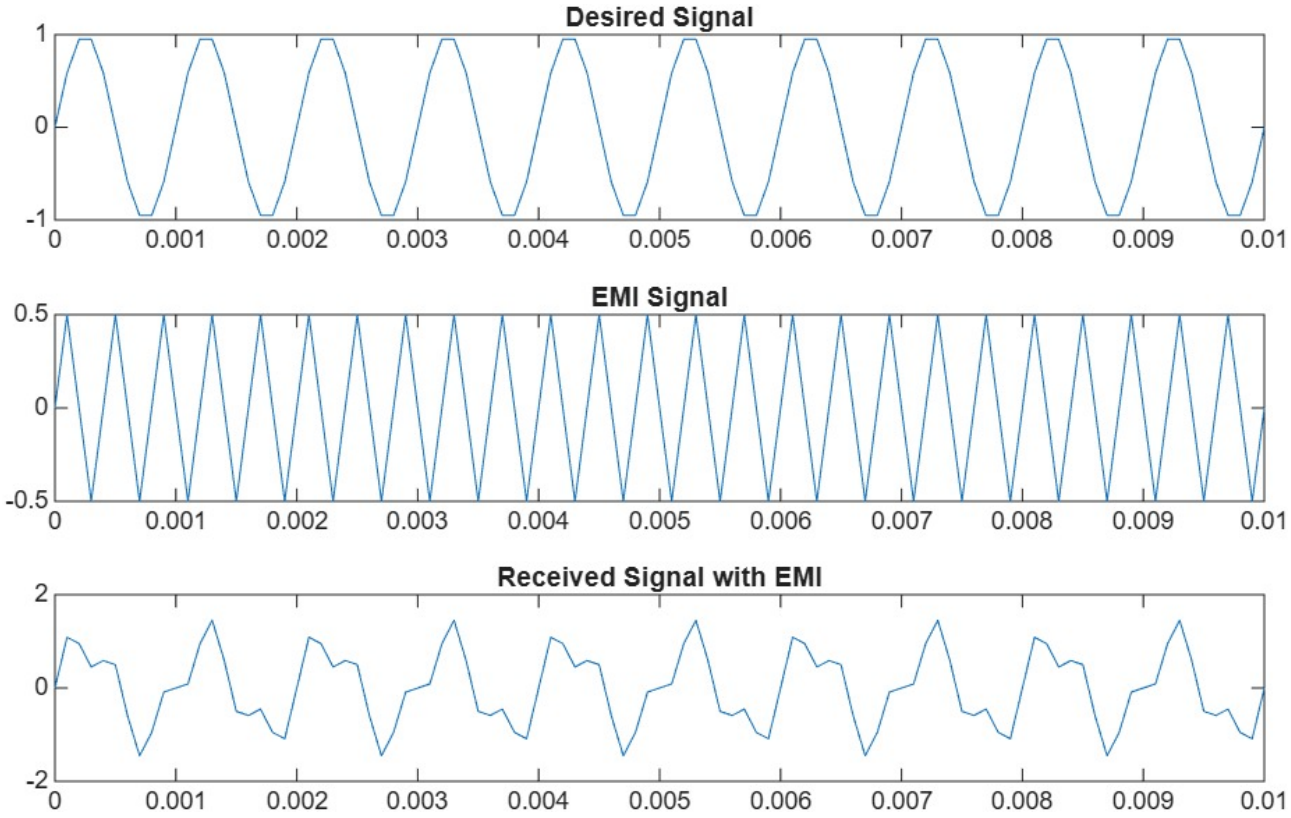


Fig. 1: Desired signal, EMI signal and received signal with EMI

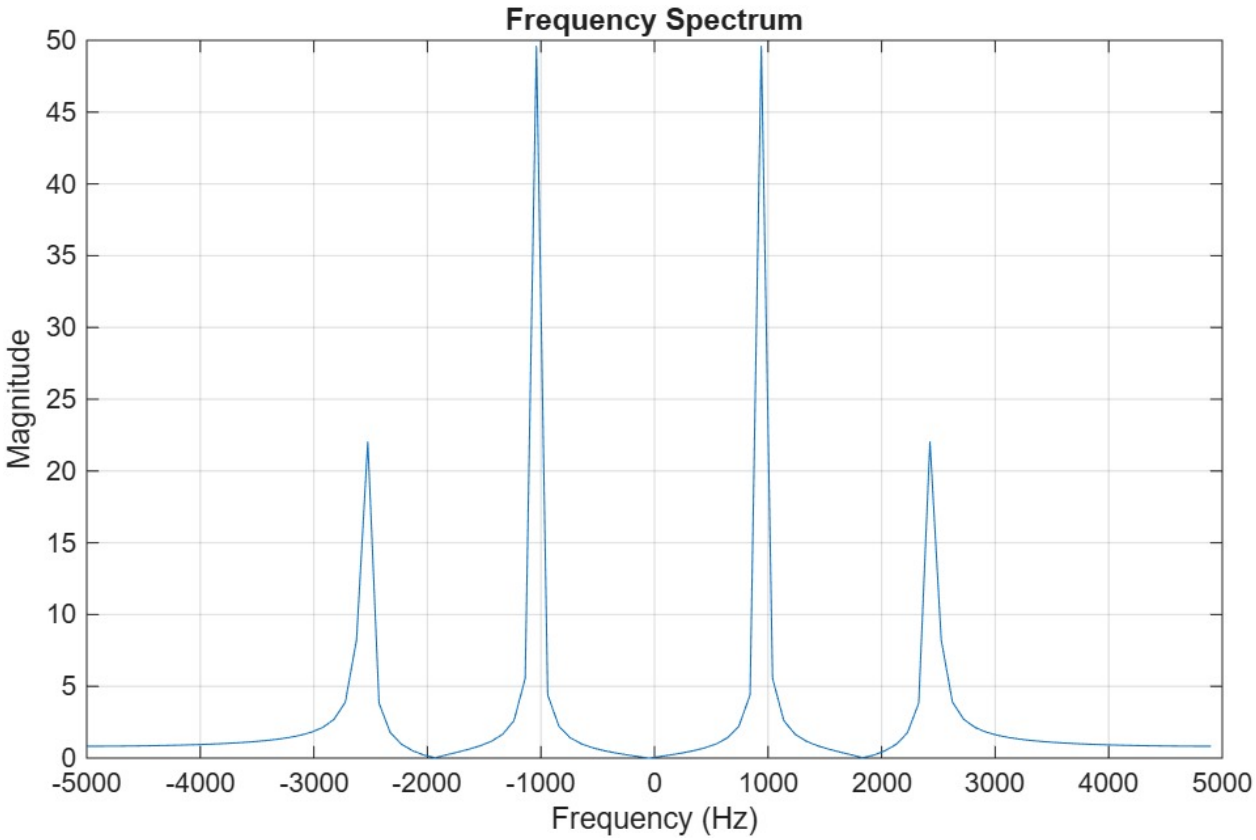


Fig. 2: Frequency spectrum of the received signal

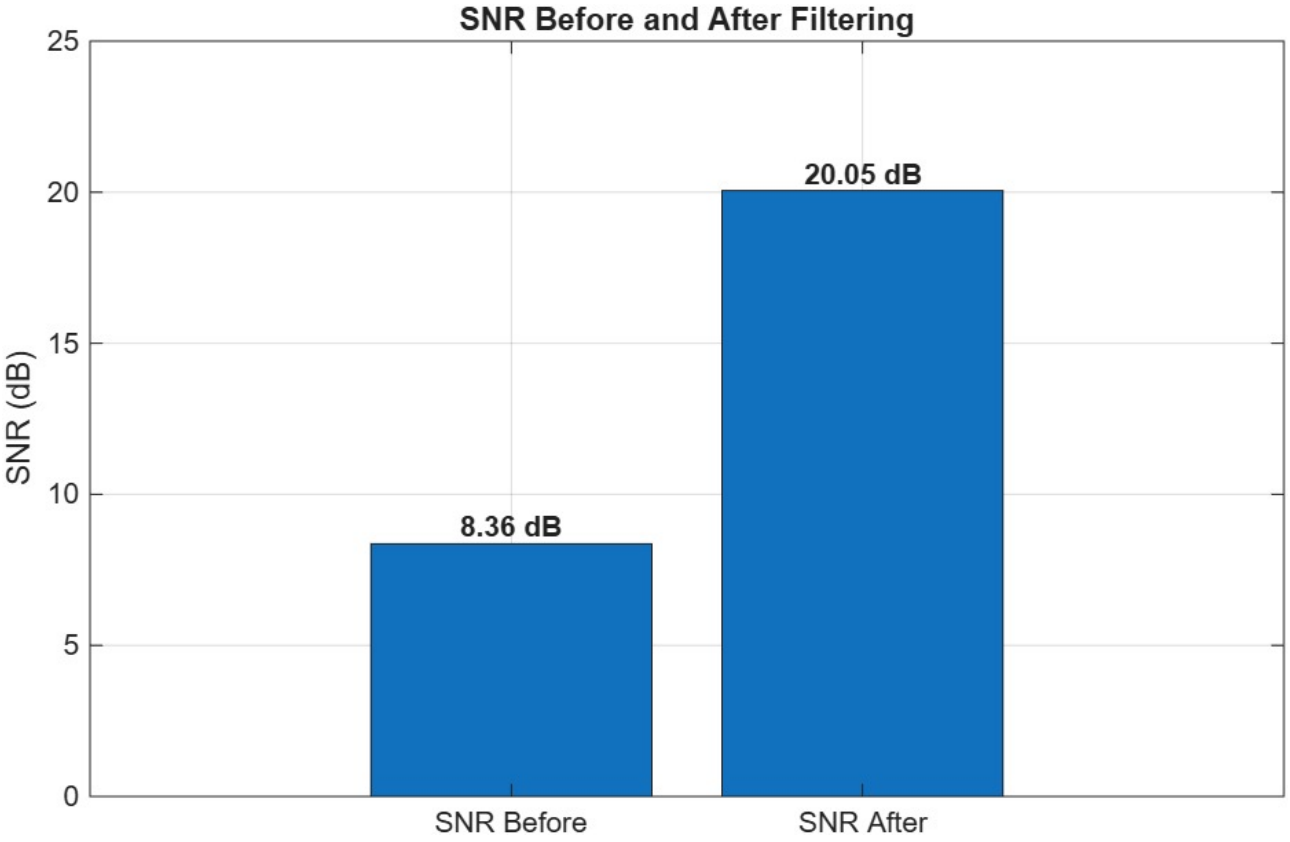


Fig. 3: SNR before and after filtering (8.36 dB to 20.05 dB)

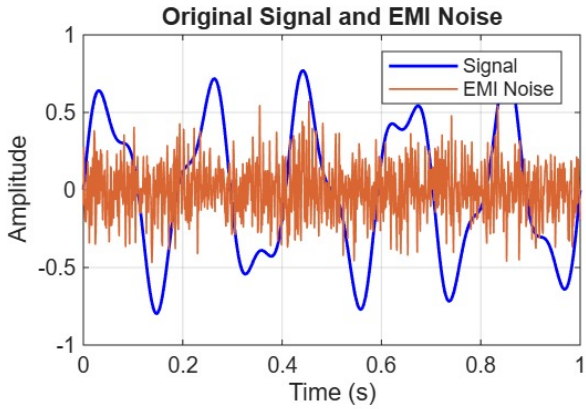
SNR Calculation

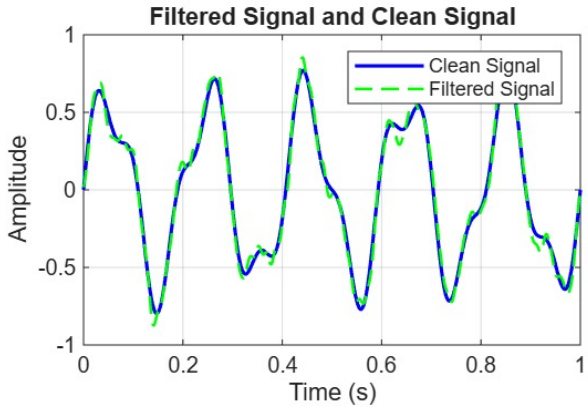
SNR Before (Pre-Filter)

Ps = mean(signal.^{2}) = 0.1998
Pn = mean(emi.^{2}) = 0.0291
SNR$_b$efore = 10*log10(Ps/Pn) = 8.36 dB

SNR After (Post-Filter)

filtered = lowpass(received, 15, Fs)
Pn$_a$fter = mean((filtered-signal).^{2}) = 0.0020
SNR$_a$fter = 10*log10(Ps/Pn$_a$fter) = 20.05 dB





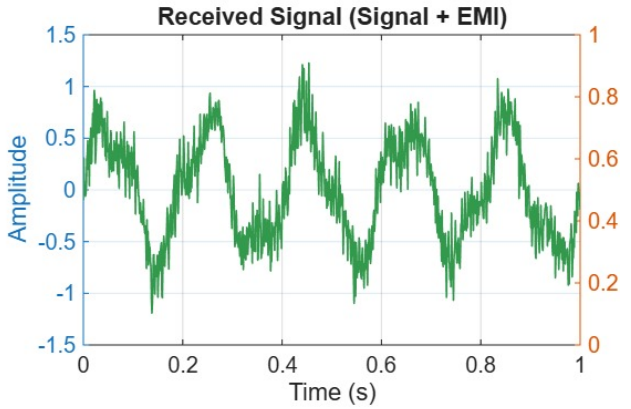


Fig. 4: SNR calculation, original signal and EMI noise, received signal and filtered vs clean signal

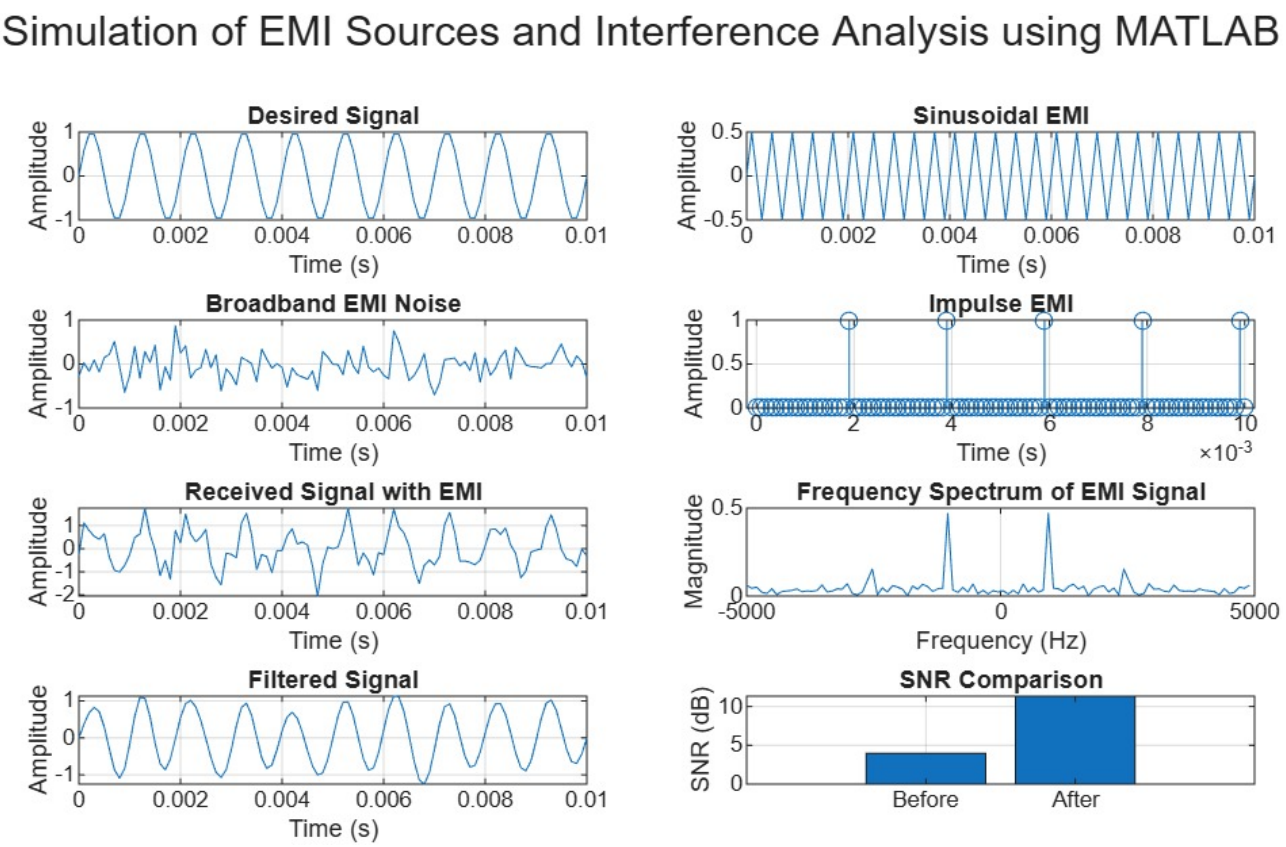


Fig. 5: Combined simulation of EMI sources and interference analysis

Result

The EMI sources were simulated and their effect on the desired signal was analysed. Low-pass filtering improved the SNR from 8.36 dB to 20.05 dB, and the filtered signal closely follows the clean signal.